

Prep Test 2: Answer Key and Rubric

75 minutes | 100 points | Keep separate from the student test.

Give follow-through credit: deduct an originating error once, not again for consistent later execution. Score missing parts separately. Do not require unrequested explanations or exact polynomial extrema.

1. Finish a piecewise graph (10 points)

1. For -2 use $2x+4$: $p(-2)=0$. At -1 and 2 use the constant branch: both values are 3 .
2. Draw $y=2x+4$ to the left of an open $(-1,2)$. Draw $y=3$ to the right of a closed $(-1,3)$. The horizontal branch includes its boundary.
3. Every real x belongs to a branch, so the domain is all real numbers. The left branch gives every y below 2 ; the right branch adds only 3 . Range: $(-\infty, 2) \cup \{3\}$.

Scoring

- 3 points: Three evaluations, 1 each.
- 2 points: Correct branch shapes and sides, 1 each.
- 2 points: Boundary coordinates and open/closed marks, 1 per boundary.
- 3 points: Domain 1; complete range including its gap 2.

2. Transform both coordinates of a given graph (12 points)

1. For source point (u,v) , the new point is $(u+2, 1-v)$: shift right 2, reflect across the x -axis, then shift up 1.
2. The vertices become $(-1,0)$, $(1,3)$, $(4,1)$, and $(6,-1)$. Join in order with straight segments and included endpoints.
3. Domain $[-1,6]$. The original range $[-2,2]$ becomes $[-1,3]$ after reflection and vertical shift.

Scoring

- 4 points: Four transformed vertices, 1 each.
- 2 points: Correctly connected straight segments. Preserve follow-through credit from an earlier coordinate error.
- 2 points: All coordinate labels 1; both included endpoints 1.
- 2 points: Domain.
- 2 points: Range.

3. Graph both composition orders with a reflection (14 points)

1. For source (u,v) , solve $1-2x=u$, so $f(g(x))$ maps to $((1-u)/2,v)$. Left-to-right vertices: $(-3/2,2)$, $(-1/2,0)$, $(1,3)$, $(2,-1)$. Domain $[-3/2,2]$.
2. For $g(f(x))$, map (u,v) to $(u,1-2v)$. Vertices: $(-3,3)$, $(-1,-5)$, $(2,1)$, $(4,-3)$. Domain $[-3,4]$.
3. Connect each vertex set in its own left-to-right order. Use solid and dashed lines as specified and mark all endpoints included.
4. For $f(g(x))$, reflect horizontally, compress horizontally by $1/2$, then shift right $1/2$. For $g(f(x))$, reflect vertically, stretch vertically by 2, then shift up 1. Equivalent correctly ordered descriptions are accepted.

Scoring

- 2 points: Two mapping rules, 1 each.
- 6 points: Four transformed vertices for each composition, 0.75 each.
- 3 points: Correct segment connections and included endpoints 2; all coordinate labels, graph names, and specified line styles 1 combined.
- 2 points: Domains, 1 each.
- 1 points: Correct transformation description for each composition, 0.5 each.

4. Evaluate and solve a composition from two graphs (10 points)

1. The first segment of f has rule $y=-2x-6$, so $f(-3.5)=1$. Then $g(f(-3.5))=g(1)=2$. Also $g(-3.5)$ is undefined because g has domain $[-3,3]$, so $f(g(-3.5))$ is undefined.
2. To solve $g(f(x))=2$, the graph of g gives inner values -1 and 1 . Solve $f(x)=-1$ or $f(x)=1$ on every segment.
3. For $f(x)=-1$, the solutions are -2.5 , -1 , and 3 . For $f(x)=1$, the solutions are -3.5 and 1 . The complete set is $\{-3.5,-2.5,-1,1,3\}$. All five inner values are in the domain of g .
4. The first segment of f has rule $y=-2x-6$, giving zero -3 . The middle segment gives zero 0 , and the last gives zero 2 .
5. f increases on $(-2,1)$ and decreases on $(-4,-2)$ union $(1,4)$. Turning-point x -coordinates, not the zeros, are interval boundaries.

Scoring

- 2 points: First nested evaluation: intermediate and final value, 0.5 each. Second: identifies undefined inside value and undefined composite, 0.5 each. Preserve valid follow-through credit.
- 4 points: Identifies both inner target values, 1; five correct x -values, 0.5 each; complete list with no extras, 0.5. Do not deduct repeatedly for one propagated graph-read error.
- 2 points: All three zeros: 0.5 each, plus 0.5 for a complete list with no extra zeros.
- 2 points: Increasing interval 1; two decreasing intervals 0.5 each. Award half of the interval credit when direction and one endpoint are correct.

5. Compare two algebraic compositions (6 points)

1. $A(B(x)) = (2x-1)^2 - 2(2x-1) = 4x^2 - 8x + 3$.
2. $B(A(x)) = 2(x^2 - 2x) - 1 = 2x^2 - 4x - 1$.
3. Equality requires $2x^2 - 4x + 4 = 0$, or $(x-1)^2 + 1 = 0$. A real square cannot equal -1 , so there are no real solutions. Equivalently, the quadratic discriminant is negative.
4. The difference $A(B(x)) - B(A(x)) = 2(x-1)^2 + 2$ is always positive, so the first composition is always greater. This observation is a valid justification, not an extra student requirement.

Scoring

- 2 points: $A(B(x))$: full substitution 1; correct expansion 1.
- 2 points: $B(A(x))$: full substitution 1; correct simplification 1.
- 1 points: Correct equality equation and valid real-solution reasoning. Preserve algebraic follow-through from one earlier error.
- 1 points: Explicitly concludes no real solutions.

6. Show a flattened crossing in a fifth-degree polynomial (14 points)

1. $P(x) = a(x+2)^2(x-1)^3$. From $P(0) = -8$, $-4a = -8$, so $a = 2$.
2. $P(x) = 2(x+2)^2(x-1)^3$ has positive leading coefficient and odd degree: left end down, right end up. Touch the axis at $(-2, 0)$ without changing sign; cross with horizontal flattening at $(1, 0)$. Label $(0, -8)$.
3. The squared factor is positive except at -2 ; the cubic factor is negative below 1 and positive above 1 . $P > 0$ on $(1, \infty)$. $P < 0$ on $(-\infty, -2)$ union $(-2, 1)$. Exclude both zeros from strict sign intervals.

Scoring

- 4 points: Multiplicity factors 2; leading-factor calculation and final equation 2.
- 1 points: Correct labeled y-intercept.
- 3 points: Both labeled zeros 1; touch at -2 1; clearly flattened crossing at 1 1.
- 2 points: Both end directions, 1 each.
- 4 points: Positive interval 1; negative intervals 2; explanation of sign change/no change 1.

7. Keep all restrictions in a composed quotient (8 points)

1. $H(x) = \sqrt{9-x^2} / (4-x^2)$.
2. The square root requires $9-x^2 \geq 0$, giving $-3 \leq x \leq 3$. The denominator requires $4-x^2 \neq 0$, so exclude -2 and 2 .
3. Domain $[-3, -2)$ union $(-2, 2)$ union $(2, 3]$. At 3 , $f(g(3)) = \sqrt{0} = 0$ and the denominator is -5 , so it is allowed. At 2 , $g(2) = 5$, so the original denominator is zero and it is not allowed.

Scoring

- 2 points: Correct substitution in numerator 1 and denominator 1.
- 2 points: Radical inequality and bounded restriction.
- 1 points: Both denominator exclusions, 0.5 each.
- 2 points: Final interval intersection including the allowed endpoints.
- 1 points: Boundary checks and explanations, 0.5 each.

8. Sketch a quartic with crossings and a touch (8 points)

1. Degree 4, leading coefficient -1. Zeros -2 and 3 are simple; 1 has multiplicity 2. $R(0) = -(2)(1)(-3) = 6$, so label (0,6).
2. Both ends fall. Cross upward at (-2,0), stay above the axis, touch and turn at (1,0), stay above until crossing downward at (3,0). Draw a smooth graph with a peak on each side of the touch; exact peaks are not requested.
3. $R > 0$ on $(-2, 1)$ union $(1, 3)$; $R < 0$ on $(-\infty, -2)$ union $(3, \infty)$. The even root does not change sign but is excluded from the strict-positive intervals.
4. Even degree makes the signs at the two far ends the same; the negative leading coefficient makes both ends fall. Odd degree makes the two ends opposite.

Scoring

- 1 points: Degree/leading coefficient 0.5; zeros and multiplicities 0.5.
- 1 points: Calculated and labeled y-intercept.
- 2 points: Labeled zeros 0.5; crossing/touch behavior 1; correctly signed smooth connections 0.5.
- 1 points: Both ends down.
- 2 points: Complete positive and negative interval lists, 1 each.
- 1 points: Even versus odd degree end-behavior explanation.

9. Use a composite model to meet a budget (10 points)

1. Use $C(A(w))$: A converts width in feet to area in square feet; C converts area to dollars.
2. $A(w) = 12w - w^2 = 36 - (w-6)^2$. On $[2, 6]$, area increases from 20 to 36. $A = 30$ gives $(w-6)^2 = 6$, so the relevant breakpoint is $b = 6 - \sqrt{6}$, approximately 3.55.
3. Cost is $96w - 8w^2$ for $2 \leq w \leq 6 - \sqrt{6}$, and $60w - 5w^2 + 90$ for $6 - \sqrt{6} < w \leq 6$. Domain $[2, 6]$. Either formula gives \$240 at the breakpoint; respect the original price condition when assigning the boundary.
4. At $w = 4$, area is 32, so use the second price branch: $C(32) = 240 + 5(2) = \$250$.
5. \$260 is above the \$240 breakpoint price, so the maximum uses the second branch: $5A + 90 = 260$ gives $A = 34$. Then $36 - (w-6)^2 = 34$ gives $w = 6$ plus or minus $\sqrt{2}$. Only $6 - \sqrt{2}$ is in $[2, 6]$.
6. The greatest width is $6 - \sqrt{2}$ feet, approximately 4.59 feet. Area and both price branches are increasing over the permitted widths, so wider permitted beds exceed the budget. Feasible widths are $[2, 6 - \sqrt{2}]$.

Scoring

- 2 points: Correct composition 1; correct unit/order explanation 1.
- 3 points: Width breakpoint 1; two cost expressions 0.5 each; branch intervals and model domain 1.
- 2 points: Area 32 and correct price branch 1; cost \$250 1.
- 3 points: Budget equation/area 0.5; exact width 1; rounded width 0.5; branch/domain check 0.5; explains why it is the greatest width 0.5. Preserve follow-through from one earlier model error.

10. Construct and graph a square-root function (8 points)

1. The domain gives $h=1$. The upper range boundary gives $k=3$, and the function is reflected downward. Substitute $(5,-1)$: $-1=2a+3$, so $a=-2$. Rule $r(x)=-2\sqrt{x-1}+3$.
2. Starting point $(1,3)$. Three useful additional points are $(2,1)$, $(5,-1)$, $(10,-3)$. Draw a decreasing square-root curve, with included start and a rightward continuation.
3. Solve $-2\sqrt{x-1}+3=0$: $\sqrt{x-1}=3/2$, $x-1=9/4$, $x=13/4$. The x-intercept is $(13/4,0)$. The function decreases on $(1,\infty)$; domain-based endpoint-inclusive wording is also acceptable.

Scoring

- 3 points: Correct h and k 1; solves for a 1; final rule 1.
- 3 points: Included labeled starting point 1; three additional labeled points 1; correct connected curve and continuation 1.
- 2 points: Exact x-intercept 1; decreasing interval 1. Accept $(1,\infty)$ or an explicitly stated decrease over its full domain $[1,\infty)$.